Business Sales Performance Analytics

Future Interns - Data Science & Analytics Internship

Task 1 Report

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Tools Used:
Power BI, Excel, Power Query

1 Abstract

This project focuses on analyzing e-commerce business sales data using Power BI to identify revenue trends, customer behavior, top-performing products, and regional sales performance. The dashboard provides a detailed overview of business KPIs including total revenue, total orders, customer count, and product sales.

The objective of this analysis is to transform raw sales data into meaningful insights that help businesses make data-driven decisions. Interactive visualizations and filters are used to improve understanding of sales patterns and customer purchasing behavior.

2 Introduction

Business analytics plays a crucial role in understanding organizational performance and improving decision-making processes. In the e-commerce industry, analyzing customer purchases, sales revenue, and product demand helps companies optimize strategies and maximize profits.

This project presents an interactive Power BI dashboard developed for analyzing e-commerce sales performance. The dashboard highlights:

- Revenue trends over time
- Country-wise sales analysis
- Product performance analysis
- Customer distribution
- Quantity sold by unit price

The dashboard enables stakeholders to monitor business growth and identify opportunities for improvement.

3 Objectives

The main objectives of this project are:

1. To analyze overall business sales performance.
2. To identify top-selling products and categories.
3. To study customer distribution across countries.
4. To understand revenue trends over time.
5. To generate actionable business insights.
6. To create an interactive and client-ready dashboard.

4 Tools and Technologies Used

Tool	Purpose
Power BI	Dashboard creation and visualization
Microsoft Excel	Data storage and preprocessing
Power Query	Data cleaning and transformation
DAX	KPI calculations and measures

Table 1: Tools and Technologies

5 Dataset Description

The dataset contains e-commerce transactional records including:

- Invoice details
- Product descriptions
- Quantity sold
- Unit price
- Customer IDs
- Country information
- Invoice dates

The dataset was cleaned and transformed before visualization to remove missing values and ensure consistency.

6 Data Cleaning Process

The following preprocessing steps were performed:

1. Removed null and duplicate values.
2. Standardized product descriptions.
3. Converted invoice dates into proper date format.
4. Filtered invalid or cancelled transactions.
5. Created calculated measures using DAX.

7 Dashboard Overview

The dashboard contains multiple visualizations to represent sales performance effectively.

Figure 1: E-Commerce Sales Performance Dashboard

The dashboard includes:

- KPI Cards
- Revenue Trend Analysis
- Country-wise Revenue Analysis
- Customer Distribution
- Product Performance Charts
- Quantity vs Unit Price Treemap

8 KPI Analysis

The dashboard highlights the following Key Performance Indicators (KPIs):

KPI	**Value**
Total Revenue	9.75M
Total Orders	26K
Total Customers	4K
Products Sold	5M

Table 2: Business KPIs

9 Visualization Analysis

9.1 Revenue Over Time

The revenue trend chart shows fluctuations in sales across different months. Significant peaks indicate seasonal demand and promotional periods.

9.2 Revenue by Country

The United Kingdom generated the highest revenue compared to other countries, contributing the majority share of total business sales.

9.3 Customer Distribution

Most customers belong to the United Kingdom region, followed by Germany and France.

9.4 Top Selling Products

Products such as WORLD WAR items and JUMBO BAG categories showed high sales quantities, indicating strong customer demand.

9.5 Quantity by Unit Price

The treemap visualization highlights relationships between quantity sold and product pricing.

10 Key Insights

- United Kingdom is the top revenue-generating country.
- Revenue shows strong seasonal variations.
- Certain products dominate overall sales volume.
- Customer concentration is highly centered in a few regions.
- Mid-range priced products contribute significantly to total sales.

11 Business Recommendations

Based on the analysis, the following recommendations are suggested:

1. Increase inventory for high-performing products.
2. Focus marketing campaigns on top-performing regions.
3. Improve customer engagement in low-performing countries.
4. Introduce targeted discounts during seasonal peaks.
5. Analyze customer purchase patterns for personalized marketing.

12 Challenges Faced

- Handling missing and inconsistent data.
- Managing large transactional datasets.
- Designing visually balanced dashboards.
- Selecting the most effective visualizations.

13 Learnings

Through this project, the following skills were developed:

- Business analytics
- Dashboard development
- Data cleaning and preprocessing
- KPI analysis
- Insight generation
- Data visualization techniques

14 Conclusion

This project successfully analyzed e-commerce sales performance using Power BI and business analytics techniques. The dashboard provides valuable insights into sales trends, customer behavior, and product performance.

The analysis helps businesses make informed decisions, improve operational efficiency, and optimize sales strategies. The project demonstrates the importance of data-driven decision-making in modern business environments.

15 References

1. Microsoft Power BI Documentation
2. Microsoft Excel Documentation
3. E-commerce Sales Dataset
4. Future Interns Data Science & Analytics Internship Guidelines